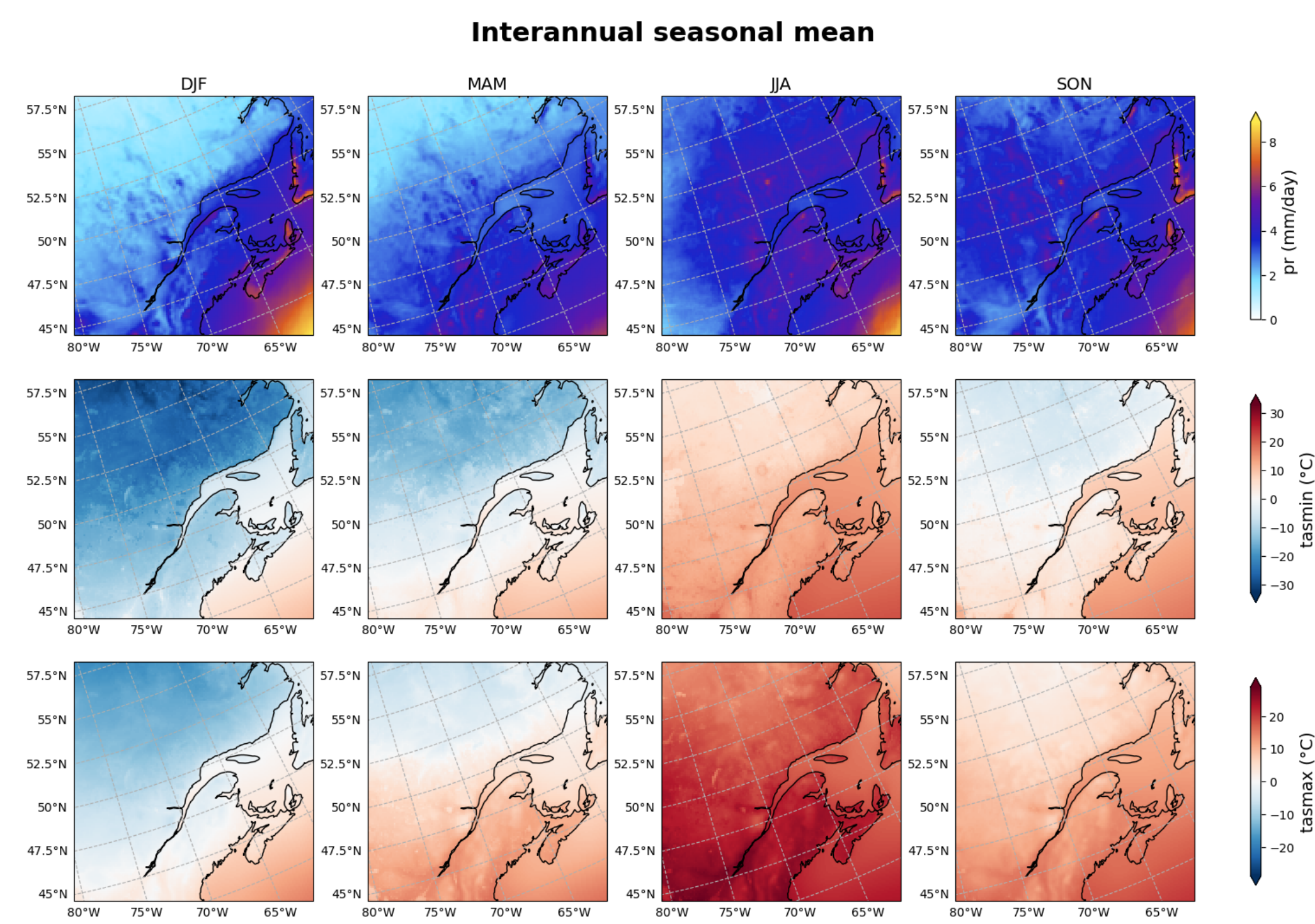


MOTIVATION AND OBJECTIVE

High-resolution climate simulations are essential for studying and mitigating the impacts of climate-related hazards. However, **Regional Climate Models (RCMs) come with prohibitive computational costs**, limiting the use of high-resolution climate data.

We explore the application of a **UNet architecture** to simultaneously downscale precipitation and temperature **without incorporating additional features**. Accurately predicting precipitation extremes and capturing fine-scale spatial patterns remain significant challenges. To address this, **we experiment with a probabilistic approach** inspired by recent advancements in generative models. The probabilistic UNet should offer the added benefit of **simplicity** while enhancing predictive performance.

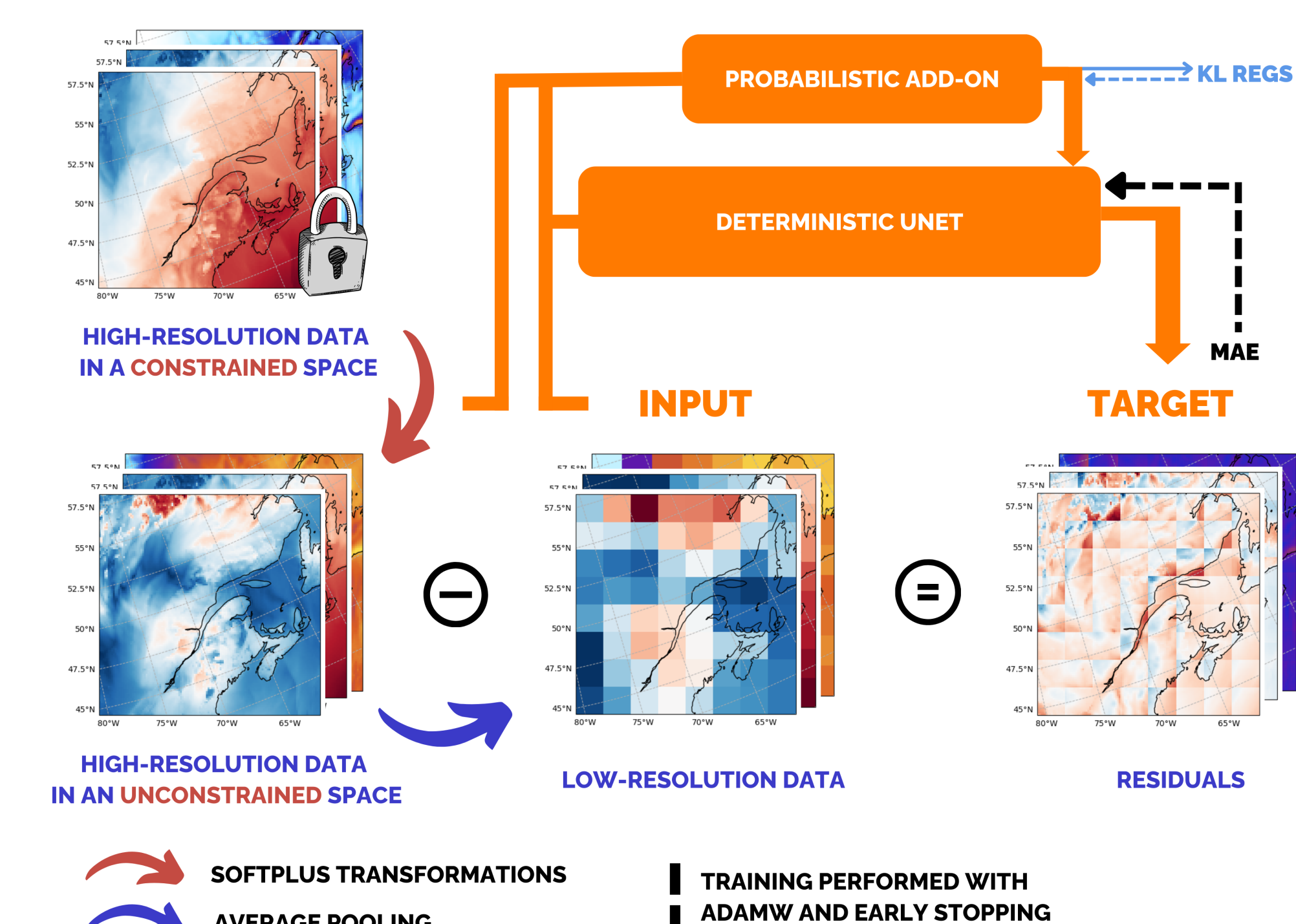
DATASET



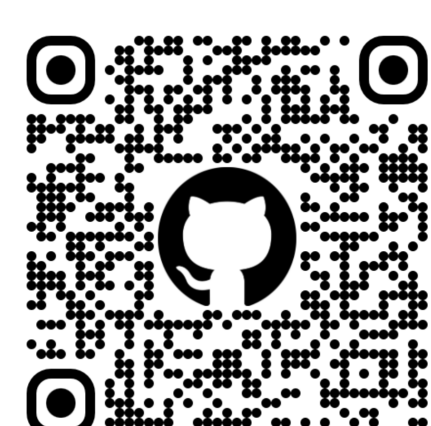
We use the Canadian Regional Climate Model Large Ensemble (CRCM5-LE), also known as **ClimEx** [1], which consists of dynamically down-scaled simulations from the global climate model **CanESM2 50-members initial-conditions large ensemble (CanESM2-LE)**.

We focus on **Southern Quebec** and Canadian Maritime provinces.

TRAINING FRAMEWORK



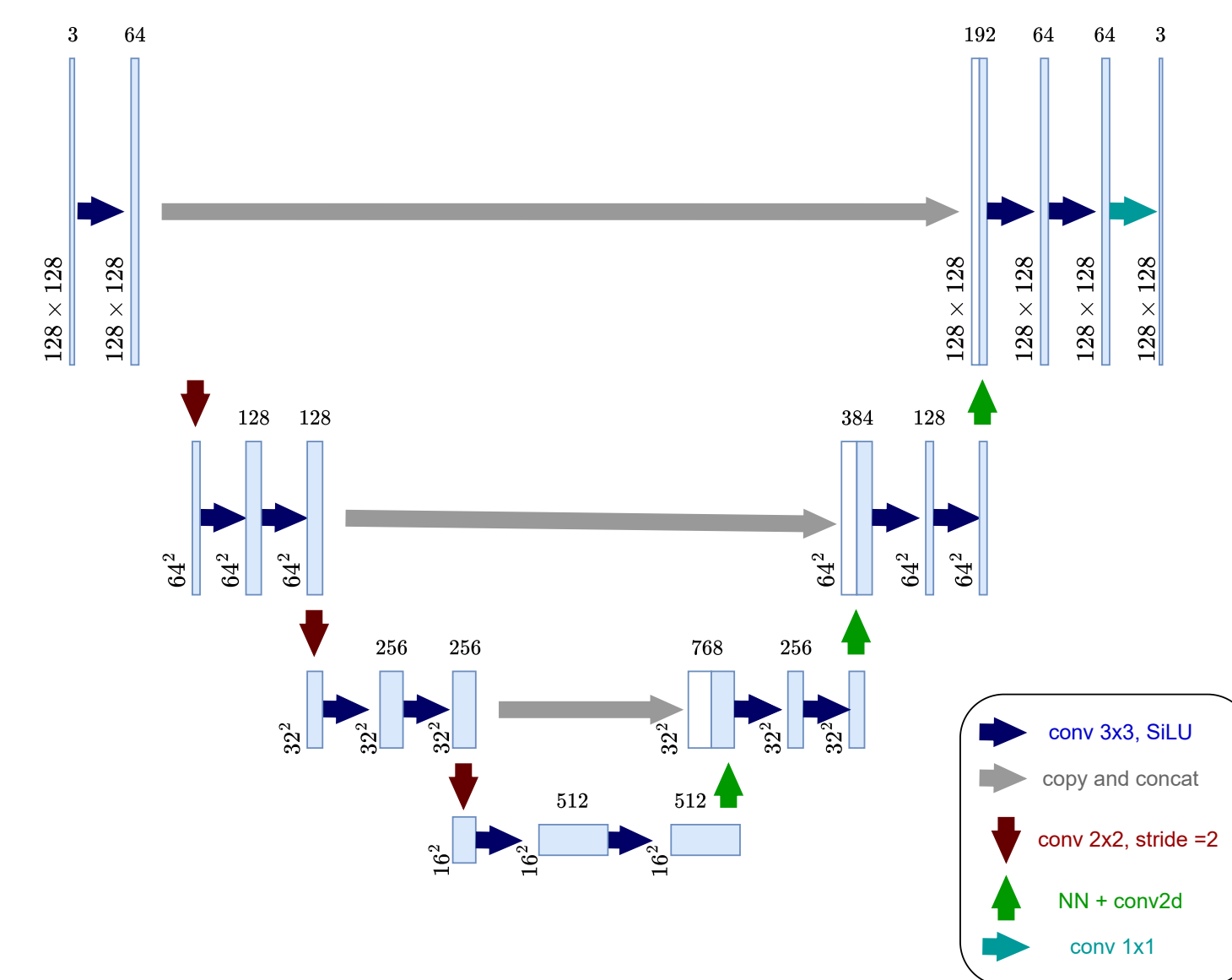
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MODELS

DETERMINISTIC UNET

- Architecture [4]: 3-level encoder-decoder with skip connections between corresponding levels.
- Input: Low-resolution data (8×8 grid) interpolated to high-resolution (128×128 grid) using nearest-neighbor interpolation.



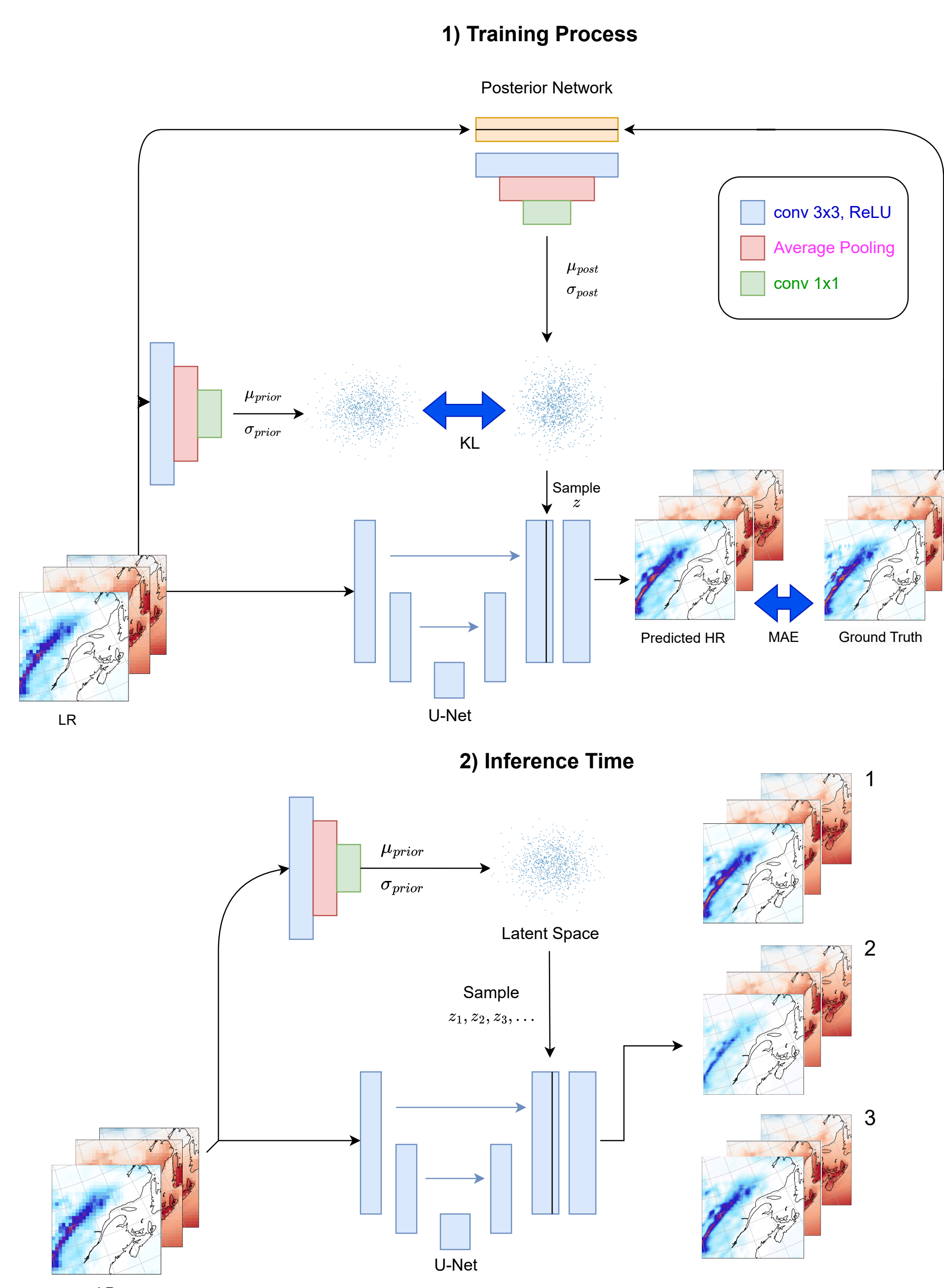
PROBABILISTIC UNET

The Probabilistic UNet [2] incorporates a variational latent space, enabling it to model multiple plausible high-resolution outputs for a single low-resolution input.

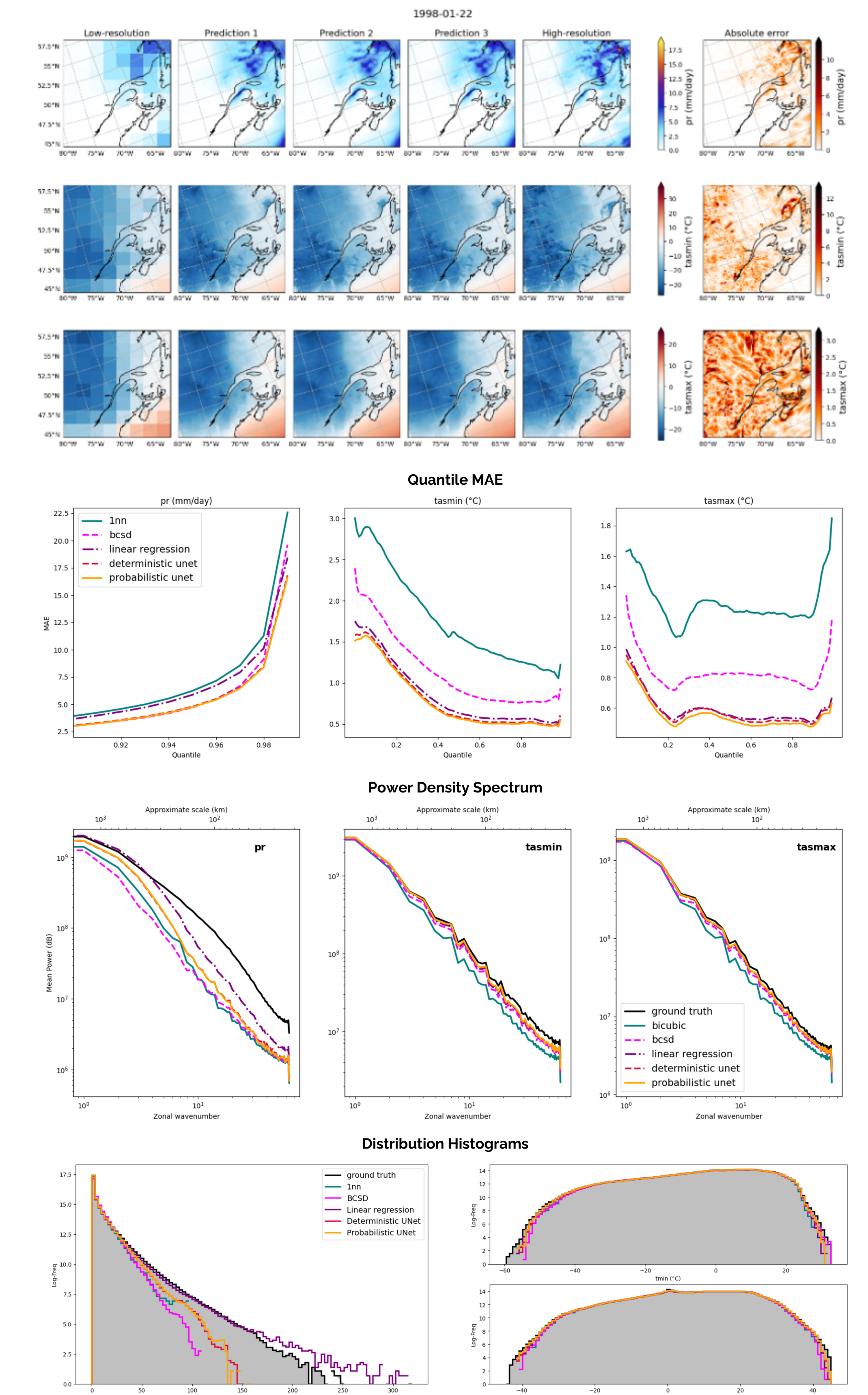
Components:

- Deterministic U-Net
- Axis-Aligned Gaussian distribution: Prior Network (input only) and Posterior Network (input + target)
- Loss Function:

$$\begin{aligned} \mathcal{L}_{ELBO} = & \text{MAE}(y, \hat{y}) \\ & + \beta_1 \cdot \mathbb{E}_{q(z|x,y)} [\text{KL}(q(z|x,y) || p(z|x))] \\ & + \beta_2 \cdot \mathbb{E}_{q(z|x,y)} [\text{KL}(q(z|x,y) || \mathcal{N}(0, I))] \end{aligned}$$



RESULTS AND FINDINGS



- Applying **softplus** transformations to train in an **unconstrained** space avoids any **incoherent** predictions and improves performance.
- Predicting the **residuals** leads to **better** results regarding point-wise metrics.
- Linear regression** poses as an **effective** and powerful comparative approach.
- Our current **probabilistic** approach **lacks** the capacity to learn and generate **diversity**.

NEXT STEPS

- Incorporate **additional input features** (elevation, humidity, wind, geopotential, land-sea mask) into our framework.
- Investigate the success of linear regression.
- Revise the probabilistic add-on and improve its capacity **through hierarchical sampling** [3].
- Explore probabilistic loss functions.

References

- M. Leduc et al. The climex project: A 50-member ensemble of climate change projections at 12-km resolution over europe and northeastern north america with the canadian regional climate model (crcm5). *Journal of Applied Meteorology and Climatology*, 58(4):663–693, Apr. 2019.
- Simon et al. A probabilistic u-net for segmentation of ambiguous images. *arXiv.org*, 2018. Accessed Dec. 27, 2024.
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- R. A. Watt and L. A. Mansfield. Generative diffusion-based downscaling for climate. *arXiv.org*, 2024. Accessed Dec. 27, 2024.